apply_call is generating an error message

https://github.com/NVIDIA/cuda-quantum/issues/1130

ROW 46

apply_call is generating an error message #1130

New issue

Closed

mmvandieren opened on Jan 23, 2024

Required prerequisites

- [x] Make sure you've read the documentation. Your issue may be addressed there.
- [x] Search the issue tracker to verify that this hasn't already been reported. +1 or comment there if it has.
- [x] If possible, make a PR with a failing test to give us a starting point to work on!

Describe the bug

I get the error message below about an invalid type being passed to the kernel, but it is a float which should be ok.

```
RuntimeError: Invalid argument type passed to kernel call (f64 != !quake.ref).
```

Steps to reproduce the bug

the code below generates the bug

```
qaoaProblem, qubit_0, qubit_1, alpha = cudaq.make_kernel(cudaq.qubit, cudaq.qubit, float)
qaoaProblem.cx(qubit_0, qubit_1)
qaoaProblem.rz(2*alpha,qubit_1)
qaoaProblem.cx(qubit_0, qubit_1)




from typing import List
# Approach 1: Define everything in one kernel

# Variable for the number of layers in the QAOA kernel
layer_count = 1 # This value can be increased, but at the cost of increased circuit depth

# Define a quantum kernel function for a parameterized quantum circuit
qaoaKernel, thetas = cudaq.make_kernel(List[float])

# Let's allocate a few qubits to our quantum circuit
# The number of qubits we'll need is the same as the number of vertices in our graph
qubit_count = 5

# Allocate n qubits to the kernel.
qreg = qaoaKernel.qalloc(qubit_count)

# Apply a Hadamard gate to each qubit
qaoaKernel.h(qreg)

# The QAOA circuit for Max Cut depends on the structure of the graph
# Each node is associated with a qubit. We'll sort the nodes and
# use the index of a vertex in this sorted list to determine the qubit
# associated with that vertex.
nodes = [2,5,7]
edges = [(2,5), (5,7)]

# Loop over the layers
for i in range(layer_count):
    # Problem unitary
    for edge in edges:
        qubitu = qreg[nodes.index(edge[0])]
        qubitv = qreg[nodes.index(edge[1])]
        # I'm getting an error message with the next line
        qaoaKernel.apply_call(qaoaProblem, qubitu, qubitv, thetas[i])
```

Expected behavior

the kernel called `qaoaProblem` should be applied to the 2 given qubits with the parameter value. The parameter value seems to generate and error message.

Assignees
No one assigned

Labels
No labels

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

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```
qaoaKernel.h(qreg)

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```

Expected behavior

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Is this a regression? If it is, put the last known working version (or commit) here.

Not a regression

Environment

- **CUDA Quantum version**: container 40733b768160
- **Python version**: 3.10.12
- **C++ compiler**: unknown
- **Operating system**: linux

Suggestions

`Type argType = otherFuncCloned.getArgumentTypes()[i];`

should be

`Type argType = otherFuncCloned.getArgumentTypes()[i++];`
the `i` was not being incremented

https://github.com/NVIDIA/cuda-quantum/blob/main/runtime/cudaq/builder/kernel_builder.cpp#L295

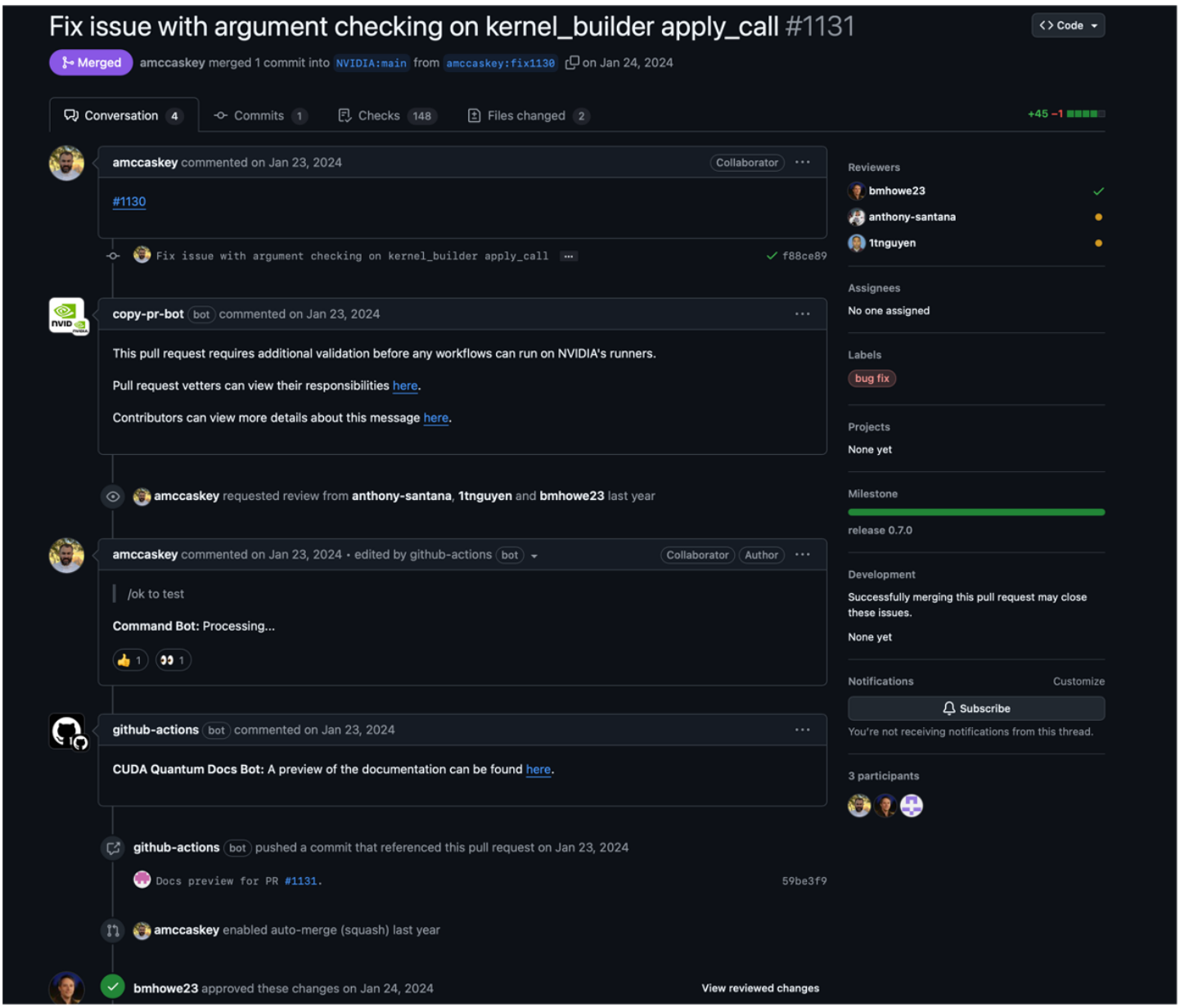
amccaskey mentioned this on Jan 23, 2024

Fix issue with argument checking on kernel_builder apply_call #1131

bmhowe23 on Jan 24, 2024 Collaborator

Fixed via #1131

bmhowe23 closed this as completed on Jan 24, 2024


Fix issue with argument checking on kernel_builder apply_call #1131
Code
Merged
amccaskey merged 1 commit into NVIDIA:main from amccaskey:fix1130 on Jan 24, 2024
Conversation 4
Commits 1
Checks 148
Files changed 2
+45 −1
amccaskey commented on Jan 23, 2024
Collaborator
#1130
Fix issue with argument checking on kernel_builder apply_call
f88ce89
copy-pr-bot bot commented on Jan 23, 2024
This pull request requires additional validation before any workflows can run on NVIDIA's runners.
Pull request vetters can view their responsibilities here.
Contributors can view more details about this message here.
amccaskey requested review from anthony-santana, 1tnguyen and bmhowe23 last year
amccaskey commented on Jan 23, 2024 • edited by github-actions bot
Collaborator
Author
/ok to test
Command Bot: Processing...
1
1
github-actions bot commented on Jan 23, 2024
CUDA Quantum Docs Bot: A preview of the documentation can be found here.
github-actions bot pushed a commit that referenced this pull request on Jan 23, 2024
Docs preview for PR #1131.
59be3f9
amccaskey enabled auto-merge (squash) last year
bmhowe23 approved these changes on Jan 24, 2024
View reviewed changes
Reviewers
bmhowe23
anthony-santana
1tnguyen
Assignees
No one assigned
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